
Which Combinations Are Worth Running? Certified Triage of Combinatorial Perturbation Screens with Convex-Cone Emergence Certificates

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Abstract

Combinatorial perturbation screens explode super-linearly: 100 single perturbations imply roughly 5,000 pairs and 160,000 triples, and only a fraction of combinations produce cell states that the single perturbations cannot already reach in combination. We ask two questions a screening team faces directly—which unmeasured combinations are worth running, and which measured ones are genuinely *emergent* rather than additive noise—and answer both from measured effects alone, with a certificate rather than a prediction. We treat a screen’s single-gene effect vectors as a fixed dictionary and define a combination as emergent when its measured effect leaves the non-negative cone of the single effects. Non-negative least squares decides cone membership exactly; on the outside, convex duality (a Farkas/KKT separator) certifies the unreachable direction. On the Norman 2019 CRISPRa combinatorial screen (data file labelled A549; the canonical line for this study is K562), all 131 measured doubles are geometrically outside the single-gene cone, but the raw unreachable fraction is confounded by effect magnitude (Spearman -0.56 ; only a minority of doubles clear their own split-half noise floor). A noise-injection null removes the confound and certifies 129/131 emergent at $p < 0.05$, of which 35 also clear a $1.9\times$ floor; the certificate recovers classical synergy (partial Spearman 0.62 controlling for magnitude). A training-free triage score—the negative cosine between two single effects—ranks emergent combinations $2.4\times$ above base rate (raw label; $1.4\times$ on the noise-robust label) *without measuring the combination*. A self-contained learned neural baseline ties the trivial additive baseline for blind prediction (held-out co-

sine 0.896 vs. 0.897), reproducing the 2025–2026 finding that learned models do not beat additivity out-of-sample; the cone’s distinct contribution is the certificate, not accuracy—it emits a separator on all 131 outside-cone doubles where predictors emit none. The certificate and its noise-aware discipline transfer to an orthogonal screen (CaRPool-seq; Cas13d knockdown in THP-1), and the machinery generalizes to arbitrary interaction order k , validated by a reachable-from-lower-order test on real doubles and a synthetic planted-epistasis triple screen (order-3 recovery AUC 1.00 ; the triple test is synthetic because the screen measures no triples). We position the method as a certified triage instrument for combinatorial screens: it flags which combinations reach states the singles cannot, model-relative to the measured dictionary—not a proof of biological synergy.

1. Introduction

Much of experimental biology is a search for interventions that move a cell from one state to another: differentiating a stem cell, repolarizing a T cell, reverting a diseased phenotype, or—in drug discovery—combining perturbations to reprogram a pathogenic program. When single perturbations are insufficient, the field turns to *combinations*, and the design space explodes super-linearly: a library of 100 single perturbations already implies roughly 5,000 pairs and 160,000 triples. No screen can measure them all, and most combinations are not worth measuring—their joint effect is what the single effects already predict. The decisive questions are therefore triage questions: *which unmeasured combinations are worth running*, and, once a combination is measured, *is its effect genuinely emergent or merely the additive sum of its parts*. These questions are rarely asked with a checkable answer, and their economic weight is large: only about one in ten clinical programs succeeds, and many late-stage failures involve insufficient efficacy, which motivates grounding combination choice in measured perturbation data rather than inferred models (Minikel et al., 2024; Nelson et al., 2015; Harrison, 2016; Begley & Ellis, 2012;

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<github.com/MasalaKimchi/cell-state-reachability>.

Prinz et al., 2011).

The computational tools available fall into two camps that, as we document in Section 4, do not overlap. Perturbation-response models—from gene-regulatory-network simulators to single-cell foundation models—are increasingly trained on real screens but are forward predictors: given a combination they return a predicted response, and a predictor structurally always returns *some* answer, so it cannot certify that a combination reaches a state the singles cannot. Network-control and Boolean-attractor methods do reason about reachability and can carry controllability guarantees, but those guarantees are properties of a hand-built or inferred model, only as trustworthy as the network assumed. Neither camp certifies emergence from measured effects. This gap matters all the more because a wave of 2025–2026 benchmarks finds that current perturbation predictors, foundation models included, frequently fail to beat mean or linear baselines on genuinely unseen perturbations (Ahlmann-Eltze et al., 2025; Wong et al., 2025; Kernfeld et al., 2025; Atti & Subramaniam, 2025). We reproduce this directly: a self-contained learned neural baseline ties the trivial additive baseline for blind prediction of held-out doubles (cosine 0.896 vs. 0.897; Section 3.3). Accuracy, in other words, is the wrong axis on which to claim a combination is special—the field’s own evidence argues for an output that is weaker to state but falsifiable.

The convex-cone move. We take the opposite starting point from model building: we do not infer a network at all. A combinatorial perturbation screen directly measures, for each single perturbation, the effect vector it produces in expression space. We treat these single-gene vectors as a fixed dictionary and ask a geometric question of each combination: does its measured effect lie in the set of shifts the singles can produce? Two modelling choices make this a convex-geometry problem. A measured single effect may be applied with non-negative weight but not reversed into an unmeasured “negative perturbation”; and several perturbations applied together are approximated, to first order, by the sum of their single effects. Individual effect vectors may raise some transcripts and lower others. The set of directions reachable by the singles is therefore the *non-negative conic hull* of the measured single-gene effect vectors, and asking whether a combination is emergent is asking whether its measured effect lies *outside* that model-relative cone—a question non-negative least squares (NNLS) answers exactly (Lawson & Hanson, 1974). This reframing turns “is this combination special?” into a decision with a checkable answer on both sides. When the combination lies inside the cone, its effect is reachable by the singles and the NNLS weights name the reachable mixture. When it lies outside, convex duality supplies a separating hyperplane—a Farkas/KKT certificate (Farkas, 1902; Kuhn & Tucker,

1951)—whose full vector certifies that no non-negative mixture of the singles reaches that direction.

The honest bar. Geometric “outside” is necessary but not sufficient. Because the unreachable residual is a normalized quantity, a low-magnitude combination measured with the same absolute noise as a large one shows an inflated apparent residual: on the Norman screen the raw unreachable fraction correlates with effect magnitude at Spearman -0.56 , and only a minority of doubles clear their own split-half noise floor. We therefore make the certificate noise-aware: a per-gene noise-injection null asks whether a reachable target, corrupted by the measured noise, would look as far outside the cone as the observed combination does. This removes the magnitude confound (the noise-aware score correlates with magnitude at $+0.14$) and yields a two-bar verdict—a permissive significance bar and a stricter effect-size bar—that a learned predictor, emitting a point estimate with no null, cannot supply.

Scope. The method is a decision instrument, not an oracle of biological truth: it certifies emergence *relative to the measured single-gene dictionary* under an explicit additivity model, and its verdicts are hypotheses for the bench, not proofs of biological synergy. The synergistic pairs it surfaces (e.g. the CEBPA/CEBPE differentiation program) are already reported by the source screen (Norman et al., 2019) and enter here as positive controls; the novelty we claim is the measured-effect, certificate-carrying *triage decision* itself. We report the source data’s provenance verbatim: the Norman combinatorial CRISPRa file used here is labelled A549, whereas the canonical Norman 2019 line is K562, and we carry that discrepancy through every result.

Contributions. This paper makes four contributions.

1. **A certified emergence test for combinatorial screens.** We formalize emergence of a combination as *non-membership* in the convex cone of measured single-gene effect vectors, and solve it with NNLS so that a verdict and—on emergence—a Farkas/KKT dual certificate follow from one convex program, made noise-aware by an injection null that removes the magnitude confound (Section 2).
2. **A validated headline result.** On the Norman combinatorial CRISPRa screen, all 131 measured doubles are geometrically outside the single-gene cone; the noise-aware null certifies 129/131 at $p < 0.05$ (35 also clearing a $1.9\times$ floor) and recovers classical synergy at partial Spearman 0.62 controlling for magnitude (Section 3.1).
3. **Training-free prospective triage, and why accuracy is the wrong axis.** A single scalar computed from

the two single effects—their negative cosine—ranks emergent combinations $2.4\times$ above base rate without measuring them, while a self-contained learned MLP ties the additive baseline for blind prediction yet emits no certificate on any of the 131 outside-cone doubles (Sections 3.2–3.3).

4. **Generality across order and across screens.** The machinery extends to arbitrary interaction order k —a reachable-from-lower-order test shrinks the certified doubles $40 \rightarrow 16$ on real data, and a synthetic planted-epistasis triple screen recovers order-3 emergence at AUC 1.00—and the certificate with its noise-aware discipline transfers to an orthogonal screen (CaRPool-seq, Cas13d knockdown in THP-1), where the raw-label triage rank also transfers but the magnitude-controlled component is screen-specific (Sections 3.4–3.5).

2. Methods

2.1. Data and effect vectors

The primary substrate is the combinatorial CRISPRa Perturb-seq screen of Norman et al. (2019). We report its provenance verbatim: the H5AD file used here is labelled cell type A549, whereas the canonical Norman 2019 line is K562; we carry that discrepancy through every result. After the source screen’s quality filters the data contain one non-targeting control, 152 single-gene perturbations, and 131 two-gene combinations (“doubles”) over 5,045 measured genes; every double’s two constituent singles are measured, so each double can be scored against a complete single-gene dictionary. For each perturbation we form a per-gene effect vector as the pseudobulk mean expression profile of that perturbation minus the pseudobulk mean of the control, giving a single-gene effect matrix $\mathbf{E} \in \mathbb{R}^{G \times P}$ with $P = 105$ single-gene atoms present as constituents of measured doubles, and a measured combination effect $\mathbf{d} \in \mathbb{R}^G$ for each double. All comparisons are reported as cosine similarity, so claims are directional. The identical operator is applied without modification to a second, orthogonal screen—the CaRPool-seq Cas13d RNA-targeting knockdown screen of Wessels et al. (2023) in THP-1 cells (28 single atoms, 158 doubles, 2,000 genes)—with no dataset-specific parameters tuned (Section 3.5).

2.2. Reachability as membership in a convex cone

Two assumptions determine the geometry. First, each measured CRISPRi effect vector enters a combination with fixed orientation: a negative coefficient would reverse the entire measured effect and amount to an unmeasured “anti-knockdown.” This does not mean knockdown lowers every transcript; each effect vector is signed. Second, applying a set of knockdowns together produces, to first order, the

sum of their individual effects; there is no natural notion of a “negative amount” of a knockdown. The set of cell-state shifts reachable by a non-negative combination of knockdowns is therefore the *non-negative conic hull* of the measured effect vectors,

$$\mathcal{C} = \left\{ \mathbf{E}\mathbf{w} : \mathbf{w} \in \mathbb{R}_{\geq 0}^P \right\} \subseteq \mathbb{R}^G, \quad (1)$$

a closed convex cone. For a measured combination effect $\mathbf{d}$, deciding whether it is reachable by the singles is deciding whether $\mathbf{d} \in \mathcal{C}$ under the additive model (up to a non-negative scale, since only direction is claimed). We quantify proximity by projecting $\mathbf{d}$ onto $\mathcal{C}$: the reachable component is the solution of the non-negative least-squares (NNLS) program

$$\mathbf{w}^* = \arg \min_{\mathbf{w} \geq 0} \|\mathbf{E}\mathbf{w} - \mathbf{d}\|_2^2, \quad \hat{\mathbf{d}} = \mathbf{E}\mathbf{w}^*, \quad (2)$$

solved with the Lawson–Hanson active-set algorithm (Lawson & Hanson, 1974). The *reachable cosine* $\cos(\hat{\mathbf{d}}, \mathbf{d})$ measures how much of the target’s direction the cone can express, and the residual $\mathbf{d} - \hat{\mathbf{d}}$ is the part it cannot. A target is declared reachable when its reachable cosine is both high in absolute terms and far above the null of Section 2.4; the same solve returns the support of $\mathbf{w}^*$, whose largest entries rank candidate contributors. A greedy forward selection adds at each step the knockdown giving the largest marginal gain in reachable cosine and yields the sparse reachability spectrum reported throughout. Its knee is a compact experimental panel, not a proof of the globally smallest perturbation set.

2.3. Infeasibility certificates via convex duality

The value of a feasibility oracle is that “no” is provable, not merely observed. When $\mathbf{d} \notin \mathcal{C}$, the separating-hyperplane theorem for convex cones (a form of Farkas’ lemma (Farkas, 1902)) guarantees a vector $\mathbf{y}$ with $\mathbf{E}^\top \mathbf{y} \leq 0$ and $\mathbf{d}^\top \mathbf{y} > 0$: a hyperplane with the whole cone on one side and the target strictly on the other. For Euclidean projection onto a cone, the residual $\mathbf{y} = \mathbf{d} - \hat{\mathbf{d}}$ supplies such a separator and is produced by the same solve rather than a separate computation. The Karush–Kuhn–Tucker (KKT) stationarity conditions of (2) provide the numerical proof: at the optimum, active (positive-weight) atoms have zero reduced gradient and inactive atoms have non-positive reduced gradient. We report the maximum stationarity violation as the certificate residual; across all reported solves it is at machine precision ($\approx 1.1 \times 10^{-11}$), confirming true optimality rather than early stopping. The biological reading is deliberately narrower than the mathematical one. When a measured double $\mathbf{d}$ lies outside the single-gene cone, the separator certifies that no non-negative mixture of the measured singles reproduces the double’s direction—the combination is *emergent* relative to the measured dictionary. This is a statement about the

dictionary under the additive model, not a proof of biological synergy; the two are related but not identical, and we test the relationship directly (Section 3.1).

2.4. Noise-aware certification

Geometric membership is necessary but not sufficient for a claim of emergence. Because the unreachable residual fraction $\|\mathbf{d} - \hat{\mathbf{d}}\|/\|\mathbf{d}\|$ is normalized by target magnitude, a low-magnitude combination measured with the same absolute noise as a large one exhibits an inflated apparent residual. We therefore calibrate every certificate against a *noise-injection null*. We first estimate a per-gene measurement standard error by splitting the cells of each condition into two halves and taking $SE_g = |t_g^{(1)} - t_g^{(2)}|/2$, where $t^{(h)}$ is the half- h effect estimate. Under the null hypothesis that the combination is truly reachable, its expected effect is the cone fit $\hat{\mathbf{d}}$; we draw $B = 200$ synthetic targets $\mathbf{d}^{(b)} = \hat{\mathbf{d}} + \boldsymbol{\varepsilon}^{(b)}$ with $\varepsilon_g^{(b)} \sim \mathcal{N}(0, SE_g^2)$, re-project each onto the cone, and record the null distribution of residual fractions. The emergence z -score is the standardized distance of the observed residual above this null mean, and the plus-one empirical p (a conservative $(1 + \#\{\text{null} \geq \text{obs}\})/(B + 1)$ estimate) is the fraction of null draws at least as extreme. The verdict has two bars: bar (a), $p < \alpha$ with $\alpha = 0.05$ (emergent above noise); bar (b), a floor ratio (observed residual over null-mean residual) ≥ 1.9 (emergent by a large margin relative to its own noise). A combination is *certified emergent* only when it clears both. Every certificate is deterministic given a seed, and each result carries a model-relative scope field naming the dictionary against which emergence was judged.

To confirm the certificate is not a magnitude artefact, we correlate the raw residual and the noise-aware z against effect magnitude, and against the classical non-additivity score $\|\mathbf{d} - (\mathbf{a} + \mathbf{b})\|/\|\mathbf{d}\|$ using both Spearman correlation and a rank-based partial correlation controlling for magnitude. The partial Spearman is the standard three-correlation estimator applied to the rank-transformed variables, $\rho_{ab:c} = (\rho_{ab} - \rho_{ac}\rho_{bc})/\sqrt{(1 - \rho_{ac}^2)(1 - \rho_{bc}^2)}$, where a, b, c are the ranks of the residual, the non-additivity score, and the effect magnitude.

2.5. Training-free prospective triage

Certification requires the measured combination; triage of *unmeasured* combinations must use the singles alone. Our default score is training-free: the negative cosine between the two single-gene effect vectors, $-\cos(\mathbf{a}, \mathbf{b})$, so that dissimilar singles rank as more likely to produce an emergent combination. We evaluate this score by ranking all measured doubles, taking the top k , and reporting the enrichment of true emergent pairs over the base rate, against both a raw unreachable-fraction label and the stricter noise-aware z

label. As an optional supervised alternative when a small labelled pilot exists, we fit a ridge model on standardized singles-only features (the pair cosine and a leave-pair-out additive gap) and evaluate it leave-one-out with a permutation null; we report it as a ceiling in supplement, not as the headline.

2.6. A self-contained learned baseline

To test whether a learned interaction model changes the picture, we implemented an additive-anchored neural baseline with no external machine-learning dependencies (numpy/scipy only). Single effects are reduced to a 50-component SVD basis retaining 95.6% of atom-span variance; a two-layer tanh multilayer perceptron (hidden width 32) with features $[\mathbf{a}, \mathbf{b}, \mathbf{a} \odot \mathbf{b}]$ predicts a residual added to the additive prediction $\mathbf{a} + \mathbf{b}$, trained by manual backpropagation with L-BFGS-B. It is evaluated *leave-pairs-out* on all 131 doubles—every test double’s own measurement is withheld—and compared against the additive baseline and against the cone’s in-sample reconstruction (which sees the measured double and is therefore not a blind forecast). This isolates the paper’s central comparison: blind predictive accuracy versus the ability to emit a certificate.

2.7. Generalization to interaction order k

The cone construction is order-agnostic. To ask whether an order- k combination is emergent relative to everything of lower order, we project its measured effect onto the cone spanned by all lower-order measured effects. For order 2 this is the single-gene cone; for order 3, the cone of singles and measured doubles. On real Norman data we compare two reference cones—singles only versus singles plus all other measured doubles—and report how the certified-emergent set changes. Because the Norman screen measures no triples, we validate the order-3 code path on a synthetic planted-epistasis screen with known ground truth (additive, 2-way-reducible, and 3-way-emergent triples), reporting recovery AUC and the confusion matrix of the two-bar verdict. The synthetic screen validates the machinery; it is not evidence about biological 3-way epistasis.

2.8. Software and reproducibility

The engine is implemented in `reachability.py` (convex-cone projection and Farkas/KKT certificate) with the combinatorial layer in `combicone.py` (`triage_combinations`, `certify_emergence`, `fit_triage_model`) and the learned baseline in `neural_baseline.py`; all three are numpy/scipy-only, with no deep-learning framework dependency. Every reported quantity derives from the published effect matrices of Norman et al. (2019) and Wessels et al. (2023) through this code, with a fixed pinned environment and a

one-command reproduction script; each figure is generated directly from those outputs.

3. Results

Figure 1 states the paper’s thesis as data. A learned model and a one-line additive rule predict held-out combination effects equally well, so predictive accuracy cannot be the criterion for calling a combination special. What distinguishes the convex-cone view is not accuracy but a falsifiable output: for every combination whose measured effect leaves the single-gene cone, the same convex program returns a separating hyperplane certifying the unreachable direction. The rest of this section builds that certificate, makes it robust to measurement noise, turns it into a training-free triage rule, and shows it generalizes across interaction order and across screens.

3.1. The certificate, and the noise bar it must clear

On the Norman combinatorial CRISPRa screen (data file labelled A549; canonical line K562), we build a dictionary of 105 single-gene effect vectors over 5,045 genes and project each of 131 measured doubles onto their non-negative cone (Section 2.2). Every one of the 131 doubles lies geometrically outside the cone: no non-negative mixture of the measured singles reproduces the measured double, and each carries a Farkas/KKT separator at machine precision.

Geometric “outside,” however, is not yet evidence of emergence. Because the unreachable fraction is normalized by target norm, a small-magnitude double measured with the same absolute noise as a large one shows an inflated apparent residual. The raw unreachable fraction indeed tracks effect magnitude at Spearman -0.56 ($p = 4 \times 10^{-12}$), and only a minority of doubles clear their own split-half noise floor (Fig. 2a). We therefore replace the point residual with a noise-injection null (Section 2.4): starting from the reachable fit $\hat{\mathbf{d}}$, we add per-gene Gaussian noise calibrated to the split-half standard error and ask how far outside the cone a genuinely reachable target would appear under that noise, repeating $B = 200$ times. The resulting z -score and plus-one empirical p quantify emergence above measurement noise, and—critically—no longer track magnitude (Spearman $+0.14$, $p = 0.13$; Fig. 2b).

This yields a two-bar verdict. Bar (a), the permissive significance test, is cleared by 129/131 doubles at $p < 0.05$. Bar (b), a stricter requirement that the residual exceed $1.9\times$ its own noise floor, is cleared by 35/131. We report both because they answer different questions: bar (a) asks whether a double is emergent at all above noise; bar (b) asks whether the emergence is large relative to how noisily it was measured. The most strongly certified pairs are SET+CEBPE ($z = 62$, floor ratio $3.5\times$), IRF1+SET ($z = 49$), and

MAPK1+PRTG ($z = 38$), each clearing both bars; the classical DUSP9+MAPK1 phosphatase-substrate synergy is strongly significant ($z = 19$) but sits just below the bar (b) floor ($1.8\times < 1.9\times$), a reminder that the two bars are distinct. A textbook apoptosis pair, BCL2L1+BAK1, passes bar (a) only weakly ($z = 3.2$, $p = 0.002$) and does not clear bar (b), and the mitotic pair KIF18B+KIF2C is the one headline pair that fails significance ($z = 1.6$, $p = 0.055$)—the noise bar demoting exactly the pairs whose apparent unreachability was a magnitude artefact. Reassuringly, the noise-aware certificate recovers classical non-additivity: the observed residual correlates with the standard synergy score $\|\mathbf{d} - (\mathbf{a} + \mathbf{b})\|/\|\mathbf{d}\|$ at Spearman 0.74, and at partial Spearman 0.62 after controlling for effect magnitude (Fig. 2b), so the signal is synergy, not scale.

3.2. Training-free triage of unmeasured combinations

The certificate needs the measured combination. The complementary question—which of the $\binom{n}{2}$ unmeasured combinations to run—must be answered from singles alone. We find that a single scalar predicts emergence prospectively: the negative cosine between the two single-gene effect vectors, $-\cos(\mathbf{a}, \mathbf{b})$. Pairs of dissimilar singles are more likely to produce an emergent combination (Spearman between $-\cos$ and the raw unreachable label -0.47 ; against the noise-aware label -0.37). Ranking the 131 doubles by this training-free score and taking the top 20 recovers 12 true top-quartile emergent pairs—a $2.4\times$ enrichment over the 0.25 base rate ($p = 3 \times 10^{-4}$; Fig. 3)—or 7/20, a $1.4\times$ enrichment, against the stricter noise-aware label. A supervised ridge model on a small labelled pilot (nine singles-only features, leave-one-out) reaches Spearman 0.43 (permutation $p = 0.002$), a modest ceiling that we report in supplement; the headline triage claim needs no training data at all.

3.3. Head-to-head against a learned baseline

To test whether a learned interaction model changes the picture, we built a self-contained neural baseline with no external dependencies: an additive-anchored two-layer MLP (hidden width 32, trained on a 50-component SVD of the 105 atoms retaining 95.6% of atom-span variance) that predicts a double’s effect from its two single effects, evaluated leave-pairs-out on all 131 doubles. Its blind held-out cosine is 0.896, statistically tied with the additive baseline’s 0.897 (the MLP wins on 49.6% of doubles), reproducing the 2025–2026 result that learned perturbation models do not beat additivity out-of-sample (Ahlmann-Eltze et al., 2025). The cone’s in-sample reconstruction cosine (0.937) is higher but is not a blind forecast—it requires the measured double. The head-to-head (Fig. 1, expanded in Fig. 4) makes the paper’s central point precise: on the metric the field optimizes, the methods tie; on the output that supports a decision—a per-pair certificate of unreachability—only the cone delivers,

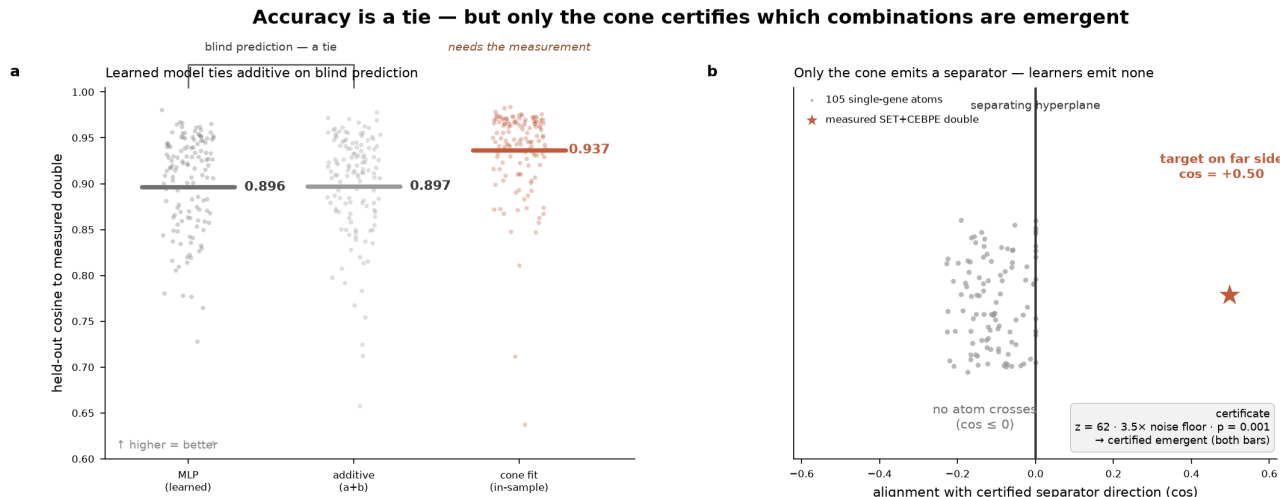


Figure 1. Accuracy is a tie—but only the cone certifies which combinations are emergent. (a) Blind (leave-pairs-out) prediction of 131 measured Norman doubles: a self-contained learned MLP (mean held-out cosine 0.896) is indistinguishable from the trivial additive baseline (0.897); the cone’s higher in-sample number (0.937) requires seeing the measurement and is not a forecast. (b) For one certified pair (SET+CEBPE), the Farkas/KKT separator places all 105 single-gene atoms on one side ($\cos \leq 0$) and the measured double strictly on the other ($\cos = +0.50$); the learned and additive predictors emit no such certificate on any of the 131 outside-cone doubles. $n = 131$ doubles; unit of replication is the perturbation pair.

emitting a valid separator on all 131 outside-cone doubles where both predictors emit none.

3.4. The certificate generalizes to interaction order k

Emergence is not restricted to pairs. The cone construction generalizes verbatim to order k : to ask whether an order- k combination is emergent *relative to everything of lower order*, we project its measured effect onto the cone spanned by all lower-order effects. On real data this sharpens the verdict. When the single-gene cone is enlarged with the other measured doubles (from 105 to 235 atoms), the number of certified-emergent doubles shrinks from 40 to 16: 24 pairs that look emergent against singles alone become reachable once other measured doubles enter the dictionary (Fig. 5a). All 131 doubles nonetheless remain geometrically outside both cones; the shrinkage is driven by effect size crossing the noise bar, not by a change in geometric membership—emergence is a property of the reference set, and the method makes that reference explicit. Because the Norman screen measures no triples, we validate the order-3 code path on a synthetic planted-epistasis screen with known ground truth: the order-3 certificate recovers planted 3-way-emergent triples from additive and 2-way-reducible ones at AUC 1.00 on the noise-aware score (0.92 on the raw residual), certifying 36/40 true emergent triples with 0/80 false positives (Fig. 5b). This is a validation of the machinery, not a claim about biological 3-way epistasis, which the data cannot support.

3.5. The certificate transfers to an orthogonal screen

Finally, we ask whether the emergence geometry is a property of one screen or of combinatorial perturbation data generally. We repeat the entire pipeline on CaRPool-seq (Wessels et al., 2023), a Cas13d RNA-targeting knockdown screen in THP-1 cells—orthogonal to Norman in perturbation modality (knockdown vs. CRISPRa activation), direction, target molecule, and cell line—with 28 single atoms and 158 measured doubles. Three findings transfer (Fig. 6). First, the magnitude confound is universal: the raw unreachable fraction tracks effect magnitude in both screens (-0.56 Norman, -0.82 CaRPool), and the noise-aware z removes it in both ($+0.14$, $+0.12$). Second, the two-bar certificate reproduces, with CaRPool showing a higher certified fraction (76/158, 48%) than Norman (40/131 under the shared pipeline). Third, the training-free triage rank transfers against the raw label ($\sim 2.2\times$ enrichment in both), but its magnitude-controlled component is screen-specific: the noise-robust triage correlation survives in Norman (Spearman 0.37) and is not significant in CaRPool (0.08). The certificate and its noise discipline generalize; the prospective triage rule generalizes in rank but should be recalibrated per screen—an honest boundary we state rather than paper over.

4. Related work

Combinatorial cell-state engineering poses an inverse-design question at scale: given a library of single perturbations, which *combinations* reach states the singles cannot,

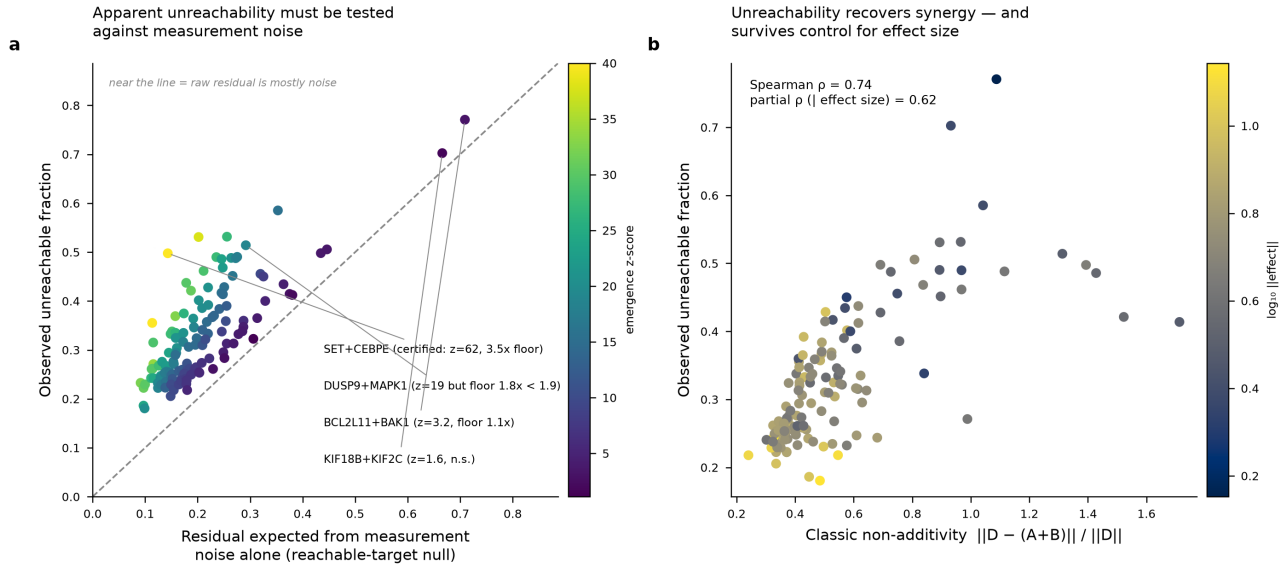


Figure 2. The emergence certificate must clear a measurement-noise bar. (a) For each of 131 Norman doubles, observed unreachable fraction (y) against the residual expected from measurement noise alone under the reachable-target null (x); points near the diagonal are consistent with noise. Colour is the noise-aware emergence z . (b) The noise-aware certificate recovers classical synergy: observed residual against the non-additivity score $\|d - (a + b)\| / \|d\|$ (Spearman 0.74; partial Spearman 0.62 controlling for effect magnitude, shown by colour). 129/131 clear bar (a) ($p < 0.05$); 35/131 also clear bar (b) ($1.9\times$ floor). $n = 131$ doubles.

and is a given combination genuinely emergent or merely additive? This is the question behind combinatorial CRISPR screens, drug-combination discovery, and multi-factor re-programming, and it is a pharmaceutical target-selection question whenever a program depends on a synergy rather than a single lever. Because the combinatorial design space explodes super-linearly, a method must also triage which of the untested combinations are worth measuring at all. Wrong target, modality, or combination choice is one contributor to the cost of drug development: only about one in ten clinical programmes reaches approval, and failure is often paid for late (Minikel et al., 2024; Wong et al., 2019). Attrition concentrates at Phase II, and roughly half of Phase II/III failures are for lack of efficacy—the wrong target or mechanism rather than safety (Harrison, 2016). The two interventions with published evidence of improving those odds both point the same way: human genetic support raises the odds of approval about 2.6-fold (Minikel et al., 2024; Nelson et al., 2015), and *measured* rather than inferred target effects provide a stronger evidence layer amid the 11–25% preclinical reproducibility documented by the Amgen and Bayer audits (Begley & Ellis, 2012; Prinz et al., 2011). A method for this problem should therefore (i) work from measured effects in the relevant screen, (ii) certify emergence rather than merely predict a combination’s response, and (iii) say so honestly, above measurement noise, when a combination is *not* emergent.

We surveyed 91 prior methods spanning 2011–2026 (92 including this work) across control-theoretic network control,

gene-regulatory-network (GRN) inference with in-silico perturbation, deep perturbation-response prediction, single-cell foundation models, and the combinatorial-screen, optimal-transport, virtual-cell, and benchmark literature that borders them. Positioning them on two axes—*what question is answered* (forward prediction versus feasibility) and *what the answer is grounded in* (an inferred or hand-built model versus directly measured effects)—exposes a gap that is exact rather than rhetorical. The methods grounded in measured effects answer only forward questions, and the methods that reason about feasibility do so only inside a model.

Model-theoretic feasibility. The most mature theory for the intervention question is network control. Structural and target controllability cast steering an assumed linear system as a matching problem over network topology (Liu et al., 2011; Gao et al., 2014), and determining-node / control-kernel methods identify driver sets from structure (Mochizuki et al., 2013; Kim et al., 2013). Boolean attractor control computes node interventions that switch a logical network between attractors (Zañudo & Albert, 2015; Zañudo et al., 2017; Joo et al., 2018; Marazzi et al., 2022), a line that remains active (Cifuentes-Fontanals et al., 2025; Metzcar et al., 2025; Murrugarra et al., 2025; Trinh et al., 2025; Del Vecchio, 2026). A data-fitted strand fits a dynamical model to omics and then applies control theory (Ronquist et al., 2017; Rukhlenko et al., 2022; Wytock & Motter, 2024), and network-influence predictors rank transcription-factor sets for a conversion (Rackham et al.,

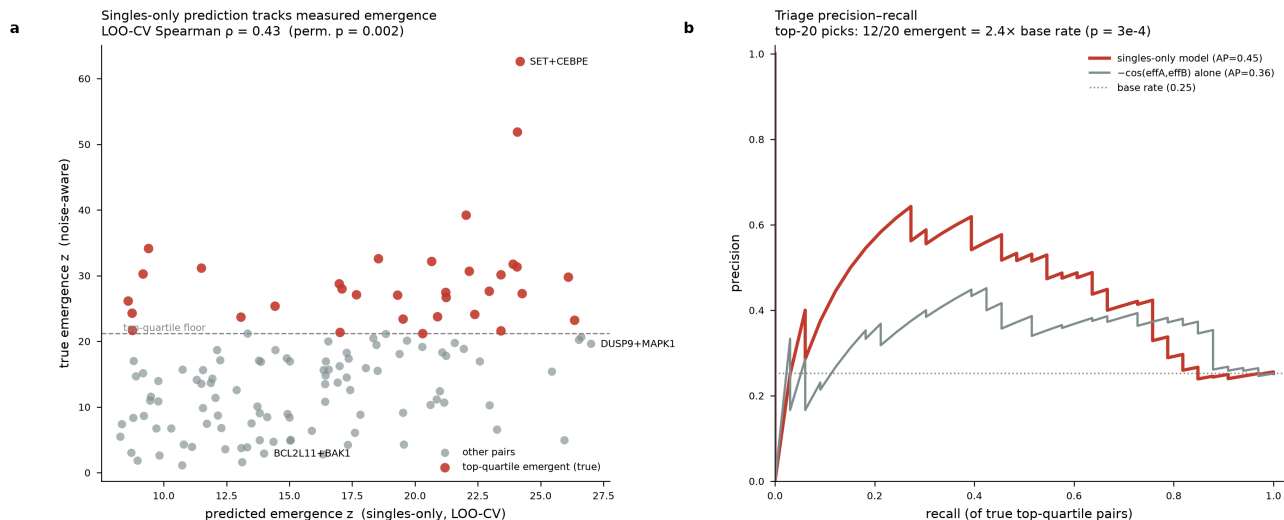


Figure 3. Training-free triage of unmeasured combinations. (a) Singles-only predicted emergence (leave-one-out) against measured noise-aware z ; Spearman 0.43 (permutation $p = 0.002$). (b) Precision–recall for flagging true top-quartile emergent pairs: the top 20 picks by the training-free score contain 12 emergent ($2.4\times$ the 0.25 base rate, $p = 3 \times 10^{-4}$). The single-scalar $-\cos(\mathbf{a}, \mathbf{b})$ alone (grey) already beats the base rate; a ridge model on a small labelled pilot (red) adds little and is reported only as a ceiling. $n = 131$ doubles.

2016; Cahan et al., 2014). These are the methods that carry a controllability or control-strategy guarantee—but the guarantee is a property of a hand-built, fitted, or inferred model, never of measured effects, and the control set is only as trustworthy as the network put in. The same dependency reappears whenever feasibility-style reasoning is imported onto a reconstructed single-cell GRN (Wang et al., 2023b) or run in a learned latent space (Han et al., 2025). The set of prior methods that emit any verdict- or certificate-like output is therefore not one family: it spans structural and Boolean network control, control-from-data, an inferred single-cell GRN, and one causally inspired perturbation predictor. What unites them is the assumed model, not the family.

Measured-grounded forward prediction. The single-cell community answers the same biological question by running an inferred network forward. CellOracle infers a GRN and simulates a transcription-factor knockout into predicted identity transitions (Kamimoto et al., 2023); related tools infer regulons, enhancer networks, or dynamic GRNs (Aibar et al., 2017; Xu et al., 2021; Bravo González-Blas et al., 2023; Wang et al., 2023a; Osorio et al., 2022), with BEE-LINE quantifying how method-dependent GRN inference is (Pratapa et al., 2020). Deep perturbation-response models are genuinely measured-grounded—they learn from real Perturb-seq data—but are resolutely forward: GEARS predicts outcomes of unseen multi-gene perturbations (Roohani et al., 2023), and compositional and disentanglement models predict responses to given perturbations (Lotfollahi et al., 2019; 2023; Hetzel et al., 2022; Piran et al., 2024). Single-cell foundation models (Theodoris et al., 2023; Cui et al.,

2024; Hao et al., 2024) and the 2025–2026 virtual-cell surge—perturbation transformers, atlas-scale models and challenges, and giga-scale perturbation atlases (Adduri et al., 2025; Roohani et al., 2025; Zhang et al., 2025; Klein et al., 2025)—extend the same forward map. Optimal-transport methods describe how cells *do* move rather than which interventions *would* move them (Schiebinger et al., 2019; Lange et al., 2022), and experimental-design methods choose the next screen (Huang et al., 2024). Every one maps perturbation to predicted response: a predictor always returns *some* answer and structurally cannot certify that a target is unreachable.

Why a feasibility verdict, not another predictor. A cluster of 2025–2026 benchmarks reports that current perturbation-prediction models, foundation models included, often fail to beat simple mean or linear baselines on genuinely unseen perturbations (Ahlmann-Eltze et al., 2025; Wong et al., 2025; Csendes et al., 2025; Kedzierska et al., 2025; Wu et al., 2025; Viñas et al., 2025; Kernfeld et al., 2025; Wei et al., 2025; Luecken et al., 2025; Atti & Subramaniam, 2025; Mao et al., 2026; Vollenweider & Bühlmann, 2026). This is the field’s own critique, and it sharpens the case for a different output: a feasibility verdict computed directly from measured effects does not extrapolate to unseen perturbations at all, and it comes with a checkable certificate and a shuffled-target null. The claim is deliberately weaker, and therefore more defensible, than predicting an unseen response.

Our position. Across all 91 prior methods (ours excluded), 14 operate directly on measured effects—and all

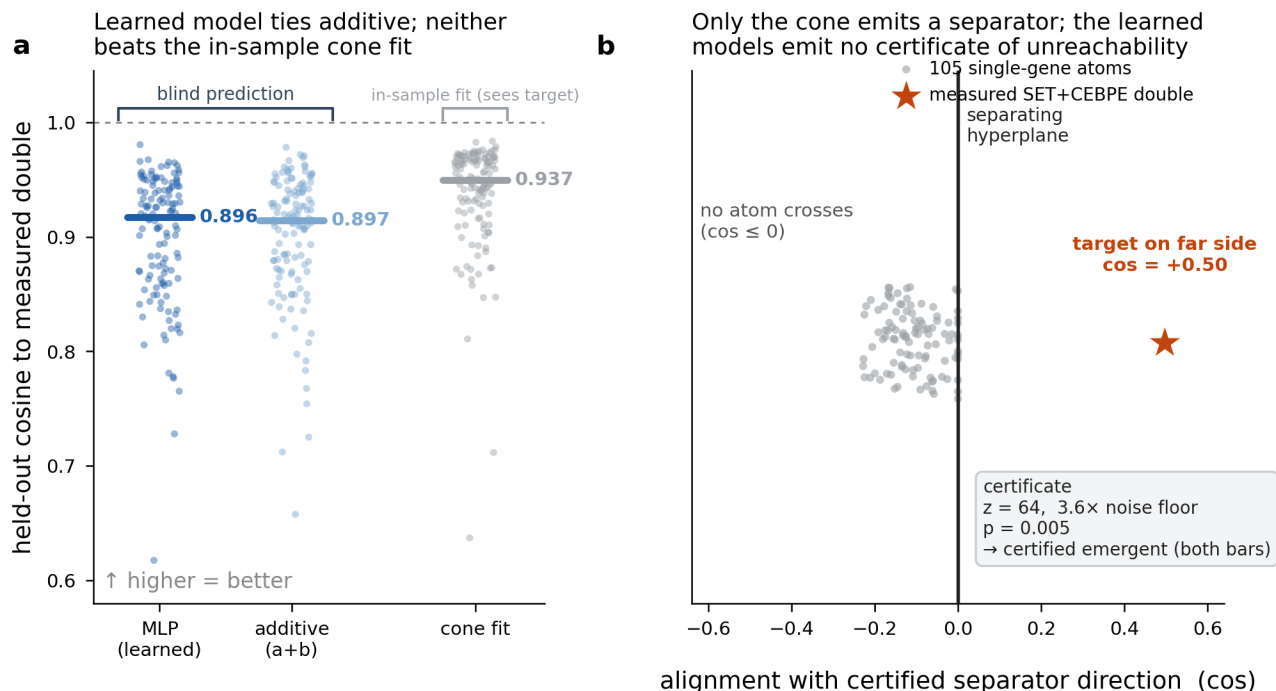


Figure 4. Head-to-head: a learned model ties additive, and neither certifies. (a) Leave-pairs-out blind prediction of 131 measured doubles: the learned MLP (mean cosine 0.896) ties the additive baseline (0.897); the cone’s in-sample fit (0.937) requires the measured double and is not a blind forecast. (b) For a certified pair, only the cone emits a separating hyperplane: all 105 single atoms fall on one side ($\cos \leq 0$) and the measured double on the other; the learned and additive predictors emit no certificate on any of the 131 outside-cone doubles. $n = 131$ doubles.

14 only predict or rank; *none* returns a feasibility verdict, emits a data-grounded certificate, or proves infeasibility, and the 13–15 partial verdict/certificate marks are all model-theoretic. The two capabilities this work combines—measured grounding and an emergence certificate—are held by *disjoint* sets of prior methods, and their intersection is empty. We compose the measured single-gene effect vectors from a combinatorial Perturb-seq screen (Norman et al., 2019) directly, with no inferred network. Because a measured single effect enters a combination with non-negative weight, the states reachable by the singles form a convex cone, and non-negative least squares (Lawson & Hanson, 1974) decides membership exactly: “inside” means the combination is additive-reachable, and “outside” is a model-relative theorem whose separating hyperplane—a Farkas/KKT certificate (Farkas, 1902)—certifies that no non-negative mixture of the singles reaches the combination’s direction. We make the certificate noise-aware, so that a magnitude-driven residual does not masquerade as emergence, and we validate it on the Norman combinatorial CRISPRa screen (all 131 doubles geometrically outside the single-gene cone; 129/131 certified above a noise-injection null, recovering classical synergy at partial Spearman 0.62). We further show that a self-contained learned model ties the additive baseline for blind prediction (Ahlmann-Eltze

et al., 2025) while emitting no certificate, that the certificate transfers to an orthogonal Cas13d screen (Wessels et al., 2023), and that a training-free score triages emergent combinations $2.4\times$ above base rate before measurement. We claim novelty for this measured-effect, certificate-carrying *emergence-triage* decision, not for the synergistic gene pairs it recovers, which the source screen (Norman et al., 2019) already reports and which we use only as positive controls.

5. Discussion

We have described a certified triage instrument for combinatorial perturbation screens. It treats a screen’s single-gene effect vectors as a dictionary and decides, by convex geometry, whether a measured combination reaches a cell-state direction that the singles cannot—answering “is this combination emergent?” with a Farkas/KKT certificate rather than a prediction, and answering “which unmeasured combinations are worth running?” with a training-free score computed from the singles alone. The value of the framework is not a new list of synergistic gene pairs—those are already in the source screens and we use them only as positive controls—but a new *kind of output*: a falsifiable, noise-aware emergence certificate computed directly from measured effects, which no forward predictor supplies.

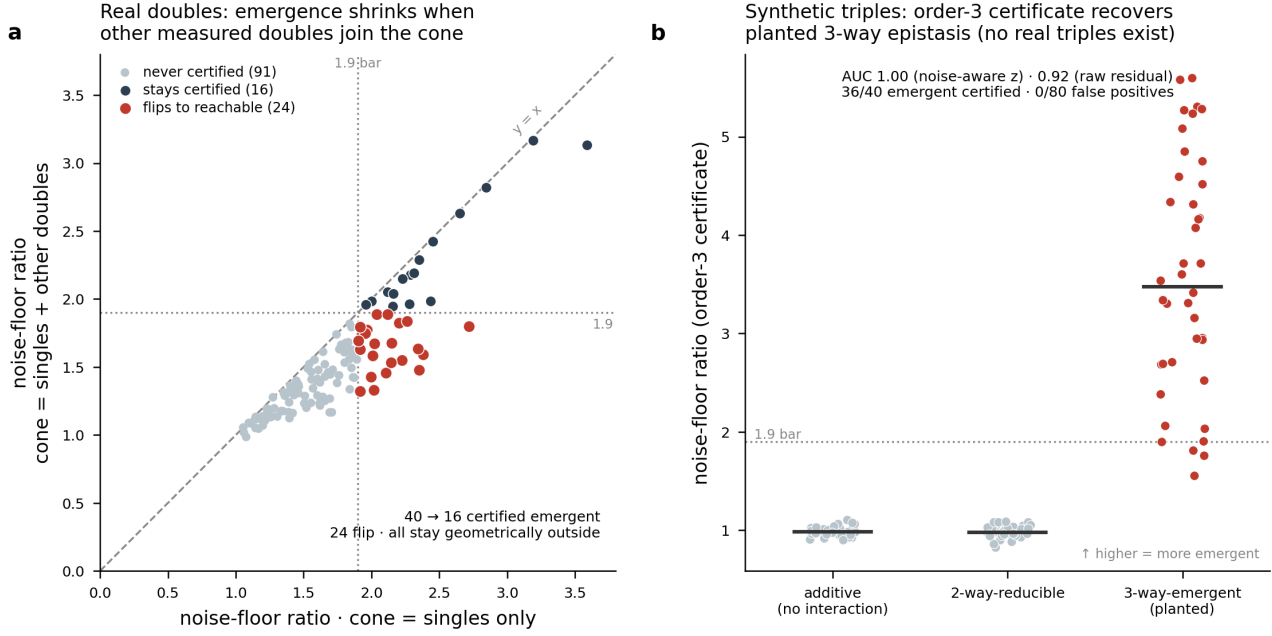


Figure 5. The certificate generalizes to interaction order k . (a) Real Norman doubles: noise-floor ratio against the single-gene cone (x) versus against the enlarged cone of singles-plus-other-doubles (y). Enlarging the reference set shrinks the certified-emergent set from 40 to 16; 24 doubles flip from emergent to reachable, while all 131 remain geometrically outside both cones. (b) Synthetic planted-epistasis triple screen (the Norman screen measures no triples): the order-3 certificate separates planted 3-way-emergent triples from additive and 2-way-reducible ones at AUC 1.00 (noise-aware z), certifying 36/40 emergent triples with 0/80 false positives.

5.1. Interpretation

The central methodological finding is that predictive accuracy is the wrong axis for claiming a combination is special. A self-contained learned model and a one-line additive rule predict held-out double effects equally well (0.896 vs. 0.897), consistent with a wave of 2025–2026 benchmarks (Ahlmann-Eltze et al., 2025; Wong et al., 2025; Kernfeld et al., 2025; Atti & Subramaniam, 2025); a screening team that ranked combinations by predicted accuracy would be ranking by a quantity on which the best available model ties a trivial baseline. The convex-cone view instead asks a decision question with a checkable answer on both sides, and only it returns a certificate: on all 131 Norman doubles the measured effect leaves the single-gene cone with a separating hyperplane at machine precision.

That geometric answer becomes trustworthy only once it clears a noise bar. The raw unreachable fraction is confounded by effect magnitude (Spearman -0.56), because a normalized residual inflates for low-magnitude combinations measured with the same absolute noise; the noise-injection null removes the confound ($+0.14$) and yields a two-bar verdict that separates “emergent at all above noise” (129/131) from “emergent by a large margin relative to its own noise” (35/131). We report both bars because they answer different questions, and because a learned point predictor, emitting no null, can supply neither. Reassuringly, the

certificate is not a measurement artefact: it recovers classical non-additivity at partial Spearman 0.62 after controlling for magnitude, so the pairs it certifies are the biologically synergistic ones.

The complement to the certificate is prospective, and here a single scalar carries the signal. The negative cosine between two single effects ranks emergent combinations $2.4\times$ above base rate without any measurement of the combination—dissimilar singles are more likely to combine into something the cone cannot reach. This is a training-free rule a team can apply to a full combinatorial catalogue before committing a single well.

5.2. Limitations

Four limitations bound the conclusion, and each maps to a concrete next step.

The most consequential is that emergence is defined *relative to the measured dictionary* under an additive model, not as biological synergy in an absolute sense. A combination we certify as emergent reaches a state no non-negative mixture of the *measured* singles reaches; if the true single-gene effect is mis-estimated, or a relevant single is absent from the library, the verdict changes. The k -way analysis makes this dependence explicit and turns it into a feature: enlarging the reference set from singles to singles-plus-doubles shrinks

the certified set from 40 to 16, because emergence is a property of the reference, and the method forces that reference to be stated.

Second, the higher-order validation is uneven. The order-2 certificate rests on real measured doubles; the order-3 certificate is validated only on a synthetic planted-epistasis screen, because the Norman screen measures no triples. The AUC 1.00 on synthetic triples establishes that the code path discriminates planted 3-way emergence from additive and 2-way-reducible combinations—it is a validation of the machinery, not evidence about real biological 3-way epistasis, which no available data can supply.

Third, generalization is partial and we state the boundary. The certificate and its noise-aware discipline transfer to an orthogonal screen (CaRPool-seq; Cas13d knockdown in THP-1): the magnitude confound and its noise-aware correction reproduce, and the two-bar certificate holds (CaRPool 48% certified). But the prospective triage rule transfers only in rank against the raw label ($\sim 2.2\times$ in both screens); its magnitude-controlled component survives in Norman and is not significant in CaRPool, so the triage score should be recalibrated per screen rather than assumed portable.

Finally, external validity is bounded by provenance and by the transcriptomic proxy. The Norman combinatorial file used here is labelled A549 whereas the canonical Norman 2019 line is K562, a discrepancy we carry through every result; and matching a target in expression space is not the same as achieving the corresponding cellular phenotype—functional confirmation remains a separate wet-lab endpoint.

5.3. Outlook

The most valuable next experiments follow directly from the limitations. Measuring a modest panel of triple perturbations would replace the synthetic order-3 validation with an in-domain one and test whether the certified-set shrinkage seen from singles to doubles continues to higher order. Applying the training-free triage prospectively—selecting the top-ranked unmeasured combinations in a fresh screen and measuring the realized emergence rate against random selection—would convert the retrospective $2.4\times$ enrichment into a forward-validated hit rate. And extending the dictionary across more screens and modalities would turn the two-screen transfer result into a generalization curve, and calibrate when the prospective score is portable. In each case the framework’s output is intended to make combinatorial screening more efficient by exposing, before the wells are run, which combinations can reach states the singles cannot.

A causal-inference agenda. Read as a causal-inference method, the certificate is a design-based counterfactual query: the single-gene effect vectors are average treatment

effects from a randomized perturbation experiment, and an emergence verdict asks whether a combination’s measured effect exceeds what any non-negative mixture of those single treatment effects produces. That framing makes the identifying assumptions explicit and turns each into a concrete robustness check. Four are worth naming. First, *interference* (SUTVA): pooled Perturb-seq lets a perturbed cell’s secreted signals alter a neighbour’s state, which would bias the single-gene effects the cone is built from; an arrayed-versus-pooled or MOI-titration comparison would bound this spillover. Second, *measurement noise as coordinated bias*: our noise-injection null models undirected per-gene noise, but a coordinated attenuation of the combination-specific signal would be a different failure mode—a sensitivity radius on the certified verdict quantifies how much coordinated bias would overturn it. Third, *construct validity*: negative-control combinations (pairs expected to be additive) should not certify, and the specificity of the synthetic screen (0/80 false positives) is the in-silico version of that check. Fourth, *guide non-compliance*: incomplete knockdown or activation attenuates the measured effect, an errors-in-variables problem an instrumental-variables treatment of guide efficiency would address. The computable half of this agenda—sensitivity radii, negative-control outcomes, and signed construct validity for the emergence certificate—is released as a causal-validation dossier accompanying the code; the wet-lab experiments above are its continuation.

Declaration of AI Usage

This work was produced with a large language model (Claude, Anthropic) operating as an agentic research system as its principal analytical instrument. Under human scientific direction, the model carried out the substantive research work: it surveyed the prior literature, formulated the convex-cone emergence-certificate framework, wrote the analysis code (`reachability.py`, `combicone.py`, `neural_baseline.py`, and the accompanying notebooks), executed the computations on the measured combinatorial perturbation screens, produced the figures and tables, and drafted and revised this manuscript. We disclose this openly and in detail because the AI’s role was central rather than incidental; a passing acknowledgement would understate it.

The human collaborator directed the project rather than the individual computations. This included setting the scientific goal (a falsifiable, screen-relative test of combinatorial emergence), selecting the primary dataset (the combinatorial CRISPRa Perturb-seq screen of Norman et al. (2019)) and the orthogonal cross-screen transfer target (Wessels et al. (2023)), steering the framing and scope, and reviewing, questioning, and validating the model’s outputs throughout. Every quantitative claim in this paper is reproducible from

the released code and data, and the numerical results were regenerated from a clean build during preparation.

The authors take full responsibility for the content of this paper, including any errors. AI-generated text and analyses were checked against the underlying data and the cited sources, and every reference was checked against a bibliographic record (OpenAlex, PubMed, arXiv, or the conference proceedings) as part of preparation; readers who find a residual error should treat it as ours to correct. We report this workflow not only for transparency but because the project is itself a case study in using AI as a scientific instrument—a way to interrogate the structure of measured biological data—and the manner of its construction is part of what it reports.

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Certified emergence geometry transfers across two orthogonal combinatorial screens

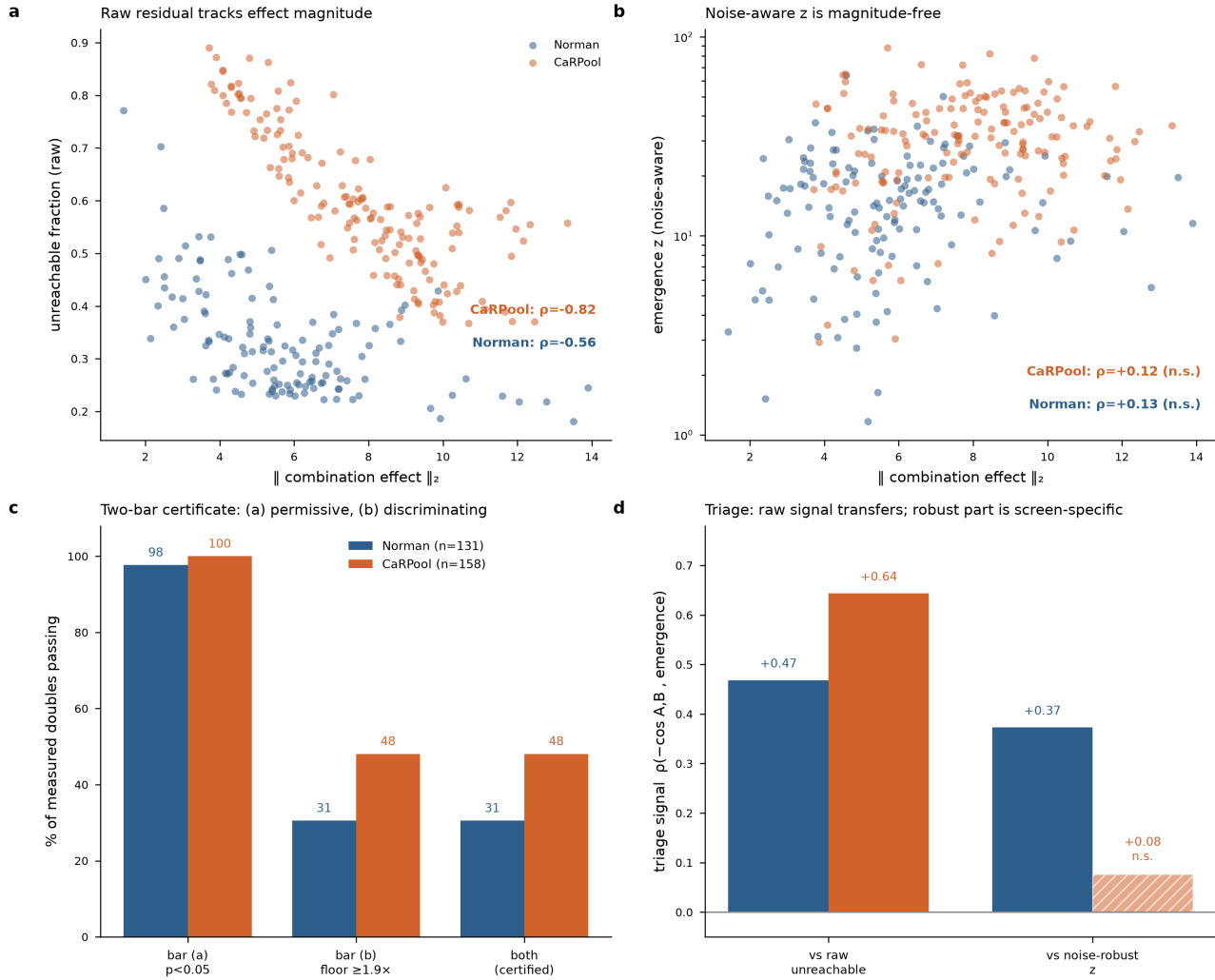


Figure 6. The certificate and its noise discipline transfer to an orthogonal screen. Norman CRISPRa doubles (blue) and CaRPool-seq Cas13d-knockdown doubles in THP-1 (orange). **(a)** Raw unreachable fraction tracks effect magnitude in both screens (-0.56 , -0.82). **(b)** The noise-aware z is magnitude-free in both ($+0.14$, $+0.12$). **(c)** The two-bar certificate reproduces (CaRPool 48% certified, Norman 31% under the shared symmetric pipeline). **(d)** The training-free triage rank transfers against the raw label ($\sim 2.2\times$ in both) but its magnitude-controlled component is screen-specific (survives in Norman, not significant in CaRPool). $n = 131$ (Norman), 158 (CaRPool) doubles.

A. Supplementary figures

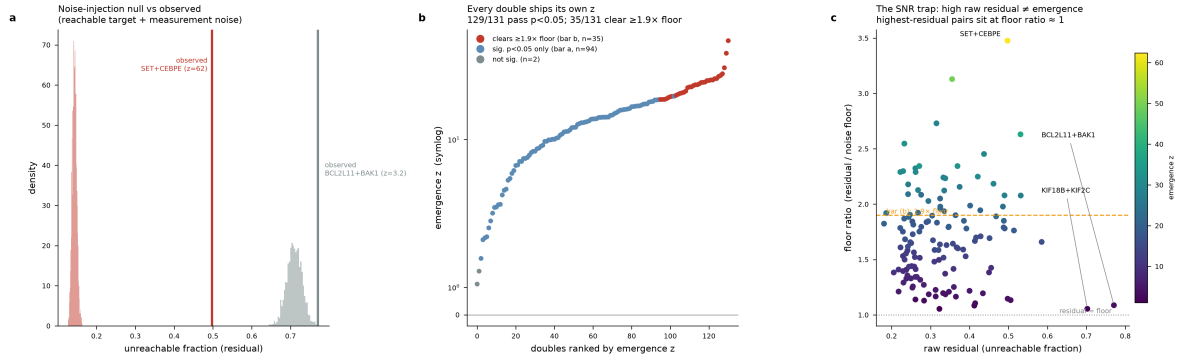


Figure 7. The noise-injection certificate in detail. For each Norman double, the observed unreachable fraction is compared against the reachable-target noise null; the bar-(a) significance call ($p < 0.05$, 129/131) and the stricter bar-(b) floor call ($1.9 \times$, 35/131) are shown, along with the “SNR trap” panel in which low-magnitude pairs show inflated *raw* residuals but collapse under the noise-aware z . This is the per-pair detail behind Fig. 2.

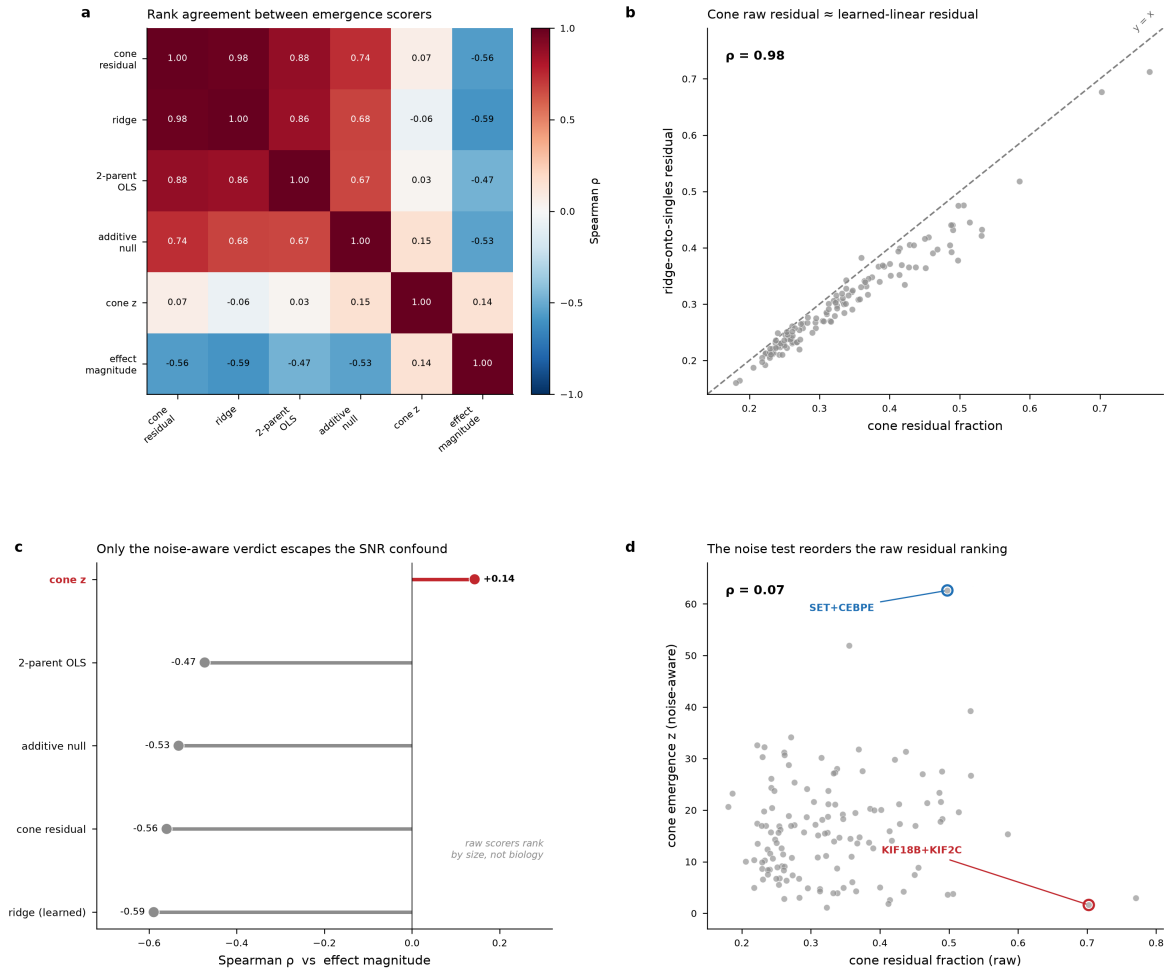


Figure 8. All raw emergence rankers are magnitude-confounded; only the noise-aware z escapes. The cone residual agrees with a ridge interaction model (Spearman 0.98), and every raw ranker—cone residual, non-additivity, ridge and OLS interaction residuals—correlates negatively with effect magnitude (-0.47 to -0.59); the noise-aware z is the only score that is magnitude-free ($+0.14$). This is the confound diagnostic behind the head-to-head of Fig. 4.

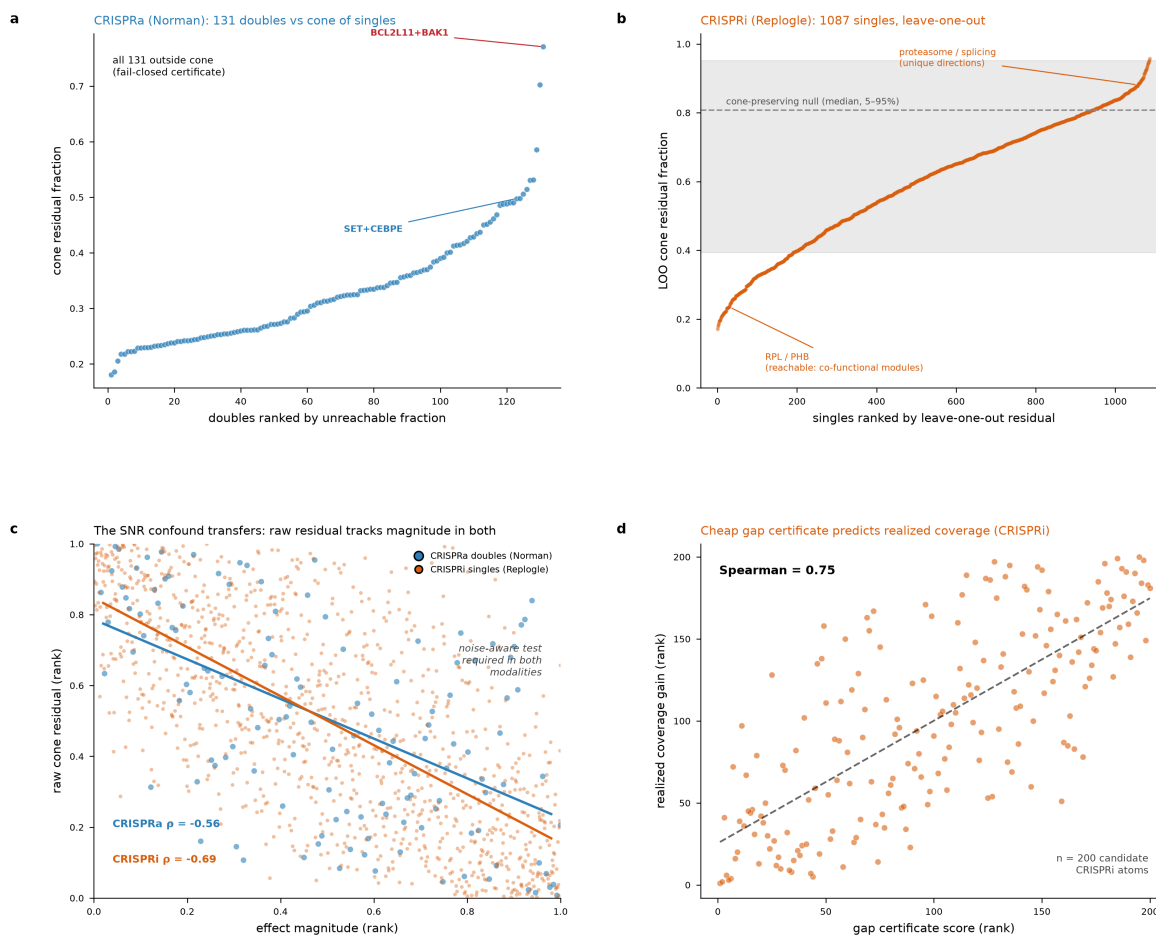


Figure 9. Cross-modality reproduction on Replogle CRISPRi. The same pipeline run on the Replogle K562 essential-gene CRISPRi screen reproduces the leave-one-out emergence spectrum and the certificate-versus-realized agreement (Spearman 0.75); the SNR/magnitude confound is present in both CRISPRa (-0.56) and CRISPRi (-0.69) and is removed by the noise-aware score in both. This is a geometry-transfer check, distinct from the orthogonal-screen certificate transfer of Fig. 6, which uses CaRPool-seq.

Reproducibility

Every number and figure in this manuscript is produced by a pinned, single-command reproduction of the released code (`reachability.py`, `combicone.py`, `neural_baseline.py`) against the published effect matrices of [Norman et al. \(2019\)](#) and [Wessels et al. \(2023\)](#). The combinatorial-triage API, its test suite, the k -way validation scripts, and a deterministic, data-free tutorial notebook accompany the code; a canonical facts sheet ties each reported value to the analysis output that produced it.